LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 4

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1. TITLE

Develop a Dynamic Web Page Using JavaScript

2. AIM

To develop a dynamic web page using HTML, CSS, and JavaScript that updates its content in real time — such as a live clock, a time-based greeting, a background colour change, and form validation — without reloading the page.

3. OBJECTIVE

The objectives of this experiment are:

To understand how JavaScript manipulates the DOM (Document Object Model).

To display a real-time clock on the web page.

To update page content dynamically based on events.

To handle events such as onclick and onsubmit.

To validate form input using JavaScript before submission.

4. THEORY

JavaScript is a client-side scripting language that runs inside the browser and allows web pages to respond to user actions without reloading.

The DOM represents the HTML page as a tree of objects; JavaScript can read and modify this tree using methods such as document.getElementById().

A dynamic web page changes its content, style, or structure after it has loaded, in response to events or timers, instead of showing only static content.

setInterval() repeatedly runs a function after a fixed time delay, which is used here to keep a clock updated every second.

Event handlers such as onclick and onsubmit let JavaScript run a function when the user interacts with a button or a form.

Form validation in JavaScript checks user input in the browser itself, giving immediate feedback before any data is submitted.

5. ALGORITHM / PROCEDURE

Create the HTML structure with a clock display, a greeting, a button, and a form.

Add basic CSS styling for the page layout and text.

Write a JavaScript function to get the current time and display it.

Use setInterval() to call this function every second, updating the clock.

Write a function that checks the current hour and displays an appropriate greeting.

Add an event handler on the button to change the background colour when clicked.

Write a validateForm() function that checks whether the name field is empty.

Display an error message dynamically if validation fails, or a welcome alert if it passes.

Open the page in a browser and test each dynamic feature.

Verify the actual output against the expected output for each test case.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Dynamic Web Page</title>
<style>
  body {
     font-family: Arial, sans-serif;
     text-align: center;
     margin-top: 50px;
     transition: background-color 0.4s ease;
  }
  #clock {
     font-size: 28px;
     margin: 20px;
  }
  #greeting {
     font-size: 20px;
     color: #333;
  }
  button {
     padding: 10px 20px;
     font-size: 16px;
     cursor: pointer;
     margin: 10px;
  }
  .error {
     color: red;
  }
</style>
</head>
<body>

  <h1>Dynamic Web Page Using JavaScript</h1>
```

```
    <div id="clock"></div>
    <div id="greeting"></div>

    <button onclick="changeColor()">Change Background Colour</button>

    <br><br>

    <form onsubmit="return validateForm()">
        <label>Enter your name:</label>
        <input type="text" id="username">
        <button type="submit">Submit</button>
        <p id="error" class="error"></p>
    </form>

<script>
    // Live clock
    function updateClock() {
        const now = new Date();
        document.getElementById("clock").innerText = now.toLocaleTimeString();
    }
    setInterval(updateClock, 1000);
    updateClock();

    // Dynamic greeting based on time of day
    function showGreeting() {
        const hour = new Date().getHours();
        let msg;
        if (hour < 12) msg = "Good Morning!";
        else if (hour < 18) msg = "Good Afternoon!";
        else msg = "Good Evening!";
        document.getElementById("greeting").innerText = msg;
    }
    showGreeting();

    // Change background colour dynamically
    function changeColor() {
        const colors = ["#f0f0f0", "#dff0d8", "#d9edf7", "#fcf8e3"];
        const random = colors[Math.floor(Math.random() * colors.length)];
        document.body.style.backgroundColor = random;
    }

    // Simple form validation
    function validateForm() {
        const name = document.getElementById("username").value;
        const error = document.getElementById("error");
        if (name.trim() === "") {
            error.innerText = "Name cannot be empty!";
            return false;
        }
        error.innerText = "";
        alert("Welcome, " + name + "!");
        return false;
    }
```

```
</script>

</body>
</html>
```

7. CODE EXPLANATION

Code	Explanation
setInterval(updateClock, 1000)	Calls updateClock() every second so the on-screen clock keeps updating.
new Date().toLocaleTimeString()	Gets the current system time and formats it for display.
new Date().getHours()	Gets the current hour, used to decide the greeting message.
document.getElementById()	Selects a DOM element so its content can be read or changed.
onclick="changeColor()"	Runs changeColor() when the button is clicked.
Math.random()	Picks a random index to choose a background colour from the array.
onsubmit="return validateForm()"	Runs validateForm() on submit; returning false stops the page from reloading.
innerText	Updates the visible text content of an element dynamically, without reloading the page.

8. TEST CASES AND EXECUTION DATA

Test Case	Action	Expected Result	Actual Result	Status
TC-01	Page loads in the browser	Live clock appears and updates every second; a time-based greeting is shown	Live clock appears and updates every second; greeting shown	Pass
TC-02	Click "Change Background Colour" button	Page background colour changes randomly	Page background colour changes randomly	Pass
TC-03	Submit form with the name field left empty	"Name cannot be empty!" message is displayed	"Name cannot be empty!" message is displayed	Pass
TC-04	Submit form with a valid name entered	Alert box welcomes the entered name	Alert box welcomes the entered name	Pass

9. OUTPUT

Output 1: Live Clock and Greeting

On loading the page, a live clock appears at the top and updates every second using setInterval(). Just below it, a greeting such as "Good Morning!", "Good Afternoon!", or "Good Evening!" is shown automatically, based on the current system hour.

Output 2: Background Colour Change

Clicking the "Change Background Colour" button runs changeColor(), which picks a random colour from a predefined list and applies it to the page background instantly, without reloading the page.

Output 3: Form Validation — Empty Field

Submitting the form with the name field empty triggers validateForm(), which displays the message "Name cannot be empty!" in red directly below the form.

Output 2: Background Colour Change

Clicking the "Change Background Colour" button runs changeColor(), which picks a random colour from a predefined list and applies it to the page background instantly, without reloading the page.

Output 3: Form Validation — Empty Field

Submitting the form with the name field empty triggers validateForm(), which displays the message "Name cannot be empty!" in red directly below the form.

Output 4: Form Validation — Valid Name

Submitting the form with a valid name instead shows an alert box that reads “Welcome, <name>!”, and the empty-field error is cleared.

10. RESULT

Thus, a dynamic web page was successfully developed using HTML, CSS, and JavaScript. The page displayed a live, continuously updating clock and a time-based greeting, changed its background colour dynamically on a button click, and validated form input in real time — all without reloading the page.